

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What is the **ONE** feature that separates **intrinsic** from **idiosyncratic** hepatotoxicity?
2. Which drug is the classic **zonal necrosis** hepatotoxin, and what is its antidote?
3. What is the single most important first step in managing any case of DILI?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 What is DILI?

Drug-induced hepatotoxicity (DILI) = an acute or chronic hepatic response to a natural or manufactured compound. **>1000 medications and herbal compounds** are known hepatotoxins (LiverTox database). DILI is the **#1 cited reason for drug withdrawal** from the market (up to 32% of withdrawals).

Risk factor	Detail
Age	Adults at higher risk than children
Sex	Females more susceptible
Race/ethnicity	African Americans — mostly trimethoprim-sulfamethoxazole; White Americans — mostly amoxicillin-clavulanate
Other	Obesity, diabetes, chronic viral hepatitis; genetic CYP polymorphisms

2 Classification

DILI is classified by **clinical presentation** (hepatocellular / cholestatic / mixed), by **mechanism** (intrinsic vs idiosyncratic), and by **histology** on biopsy. Most cases are asymptomatic; when symptomatic, **jaundice** is the most common sign. Hepatocellular injury → ↑↑ aminotransferases (ALT/AST); cholestatic injury → ↑ alkaline phosphatase (ALP).

The ONE discriminator — intrinsic vs idiosyncratic

	Intrinsic	Idiosyncratic
Dose relationship	Dose-dependent	No dose relationship
Predictability	Predictable	Unpredictable
Latent period	Brief	Variable
Mortality	Can be high	—
Classic example	Acetaminophen	Isoniazid, ketoconazole, sulfonamides, phenytoin

Idiosyncratic reactions split further — **hypersensitivity** (immunologic, onset 1–5wk, recurs with rechallenge, rash/fever/joint pain: sulfonamides, phenytoin, Augmentin) vs **metabolic** (aberrant metabolism, onset weeks–months, recurs with rechallenge: isoniazid, ketoconazole, amiodarone, phenytoin).

3 Injury pattern → drug (the core table)

Pattern	Drug example
Acute hepatocellular injury	Isoniazid · Aspirin (Reye's) · Phenytoin
Chronic hepatocellular injury	Amoxicillin-clavulanate · Valproic acid · Amiodarone
Zonal (centrilobular) necrosis	Acetaminophen

Acute cholestasis	Anabolic steroids
Chronic cholestasis	Amoxicillin-clavulanate
Steatosis (microvesicular)	Tetracycline · Valproic acid
Granulomas	Talc exposure

4 Mechanism, one line per drug (recall column)

Acetaminophen	Overdose depletes glutathione → toxic metabolite NAPQI accumulates → zonal necrosis. Antidote: N-acetylcysteine (replenishes glutathione, directly detoxifies NAPQI, improves hepatic blood flow, limits oxidative stress).
Isoniazid	Metabolised to acetylhydrazine → toxic CYP450 product → idiosyncratic hepatocellular injury; genetic susceptibility.
Aspirin (+ viral illness, esp. children)	Mitochondrial dysfunction → microvesicular steatosis = Reye's syndrome (confusion, vomiting, ↑ ammonia).
Phenytoin	Toxic CYP450 metabolite → immune-mediated hepatocellular necrosis and inflammation.
Amoxicillin-clavulanate	Clavulanic acid triggers an idiosyncratic hypersensitivity reaction → immune attack on hepatocytes/bile duct cells → cholestatic injury.
Tetracycline	Impairs mitochondrial fatty-acid oxidation → microvesicular steatosis.
Valproic acid	Impairs mitochondrial β -oxidation + oxidative stress → triglyceride accumulation (steatosis) / chronic hepatocellular injury.
Anabolic steroids (stanozolol)	Impair bile transporter proteins → intrahepatic cholestasis, usually reversible on stopping; treat with ursodeoxycholic acid.
Amiodarone	Accumulates in liver → phospholipidosis + mitochondrial dysfunction → chronic hepatocellular injury ± fibrosis, which may persist even after stopping.
Sulindac (NSAID)	Inhibits canalicular bile salt transport → cholestatic injury (itching, jaundice).
Niacin (sustained-release)	Highest hepatotoxic potential of the antihyperlipidemics → ↑ aminotransferases ± mixed cholestatic pattern. (Statins rarely cause significant injury, though asymptomatic transaminase rises are common.)
Methotrexate	↑ homocysteine → oxidative stress → activates hepatic stellate cells → fibrosis.
Sulfasalazine	Usually within the first month; can be hepatocellular or cholestatic.
Azathioprine	Nodular regenerative hyperplasia (NRH) → can cause non-cirrhotic portal hypertension.

Anti-hyperlipidemic mechanisms of hepatotoxicity (varies by drug): effects on the **cytochrome P450 system**, **impairment of bile acid transport**, an **immune-mediated inflammatory response**, and **oxidative stress** from intracellular damage.

5 Diagnosis & management

- **Hepatocellular** injury → ↑↑↑ ALT/AST. **Cholestatic** injury → ↑ALP (±GGT).
- Most DILI is asymptomatic; **jaundice** is the most common presenting sign when symptoms occur.
- **Mainstay of treatment: remove the offending agent.** Prognosis for recovery is usually favourable after discontinuation.
- Acetaminophen overdose is the exception with a specific antidote — N-acetylcysteine.

6 The 3-step mechanistic cascade

Drug or its metabolite → **(1)** direct cell stress or reactive oxygen species during CYP450 oxidation → **(2)** impaired mitochondrial function → **(3)** triggered immune reactions.

P Must-remember pearls

- **Acetaminophen** is the classic intrinsic, dose-dependent, predictable hepatotoxin — overdose is treated with **N-acetylcysteine**.
- Idiosyncratic reactions are unpredictable and **not** dose-related — split into hypersensitivity (immune, fast, rash/fever/joint pain) vs metabolic (aberrant metabolism, slower onset).
- **Amoxicillin-clavulanate** is the prototype idiosyncratic **cholestatic** hepatotoxin (via clavulanic acid).
- **Reye's syndrome** = aspirin given during a viral illness (especially in children) → microvesicular steatosis.
- DILI's biggest real-world consequence: it is the **leading cause of drug withdrawal** from the market.

Q Test yourself

1. A 6-year-old with influenza is given aspirin for fever and develops confusion, vomiting, and elevated ammonia. Diagnosis?

- | | |
|-----------------------------|------------------------|
| A. Acetaminophen toxicity | B. Reye's syndrome |
| C. Isoniazid hepatotoxicity | D. Amiodarone toxicity |

2. Which combination of features best fits INTRINSIC hepatotoxicity?

- | | |
|--|--|
| A. Unpredictable, no dose relationship, variable latency | B. Dose-dependent, predictable, brief latent period |
| C. Immune-mediated with rash/fever/joint pain | D. Recurs only with rechallenge, onset weeks to months |

3. A patient who recently finished amoxicillin-clavulanate develops jaundice, pruritus, and a markedly raised ALP with only mild ALT elevation. Most likely injury pattern?

- | | |
|----------------------------|-----------------------------|
| A. Zonal necrosis | B. Cholestatic hepatitis |
| C. Granulomatous hepatitis | D. Microvesicular steatosis |

4. Which drug carries the highest hepatotoxic potential among the antihyperlipidemics?

- | | |
|-----------------|-----------------------------|
| A. Atorvastatin | B. Sustained-release niacin |
| C. Ezetimibe | D. Fenofibrate |

5. What is the mechanism of N-acetylcysteine in acetaminophen overdose?

- | | |
|---|--|
| A. Inhibits CYP450 to prevent NAPQI formation | B. Replenishes glutathione and directly detoxifies NAPQI |
| C. Blocks acetaminophen absorption in the gut | D. Increases renal clearance of acetaminophen |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1. — B — aspirin during a viral illness in a child causing mitochondrial dysfunction/microvesicular steatosis is classic Reye's syndrome.
2. — B — intrinsic hepatotoxicity is dose-dependent, predictable, with a brief latent period (e.g. acetaminophen).
3. — B — jaundice, pruritus, disproportionate ALP rise = cholestatic pattern, the classic Augmentin injury.
4. — B — sustained-release niacin has the highest hepatotoxic potential of the antihyperlipidemic drugs.
5. — B — NAC replenishes glutathione and directly detoxifies NAPQI, also improving hepatic blood flow and limiting oxidative stress.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · from Asst. Prof. Nidhal Abdulkader & Asst. Lect. Nermin Jamal's DILI lecture
Study alongside the lecture, not instead of it.